HW5 final

Monday, 24 March 2025 2:34 PM

Problem 1

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y} = \frac{\partial^2 u}{\partial y^2} \qquad (x,y,t) \Leftrightarrow (\xi,\eta,\tau)$$

(a) We can use coordinate transformations: (using $t=\tau$ & $\tau \neq f(x,y)$)

$$\frac{\partial}{\partial x} = \xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}$$

$$\frac{\partial}{\partial y} = \xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}$$

$$\frac{\partial}{\partial \tau} = \frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}$$

term 1 term 2 term 3 term 4

$$\therefore \boxed{\left(\frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}\right)u + \left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right)u^2 + \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)uv = \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)\left(\frac{\partial u}{\partial y}\right)}$$

Chain-Rule Conservation-Law form

(b) To maintain conservation have to convert to Strong Conservation-Law form

1) $J \times$ (PDE from above)

2) Move J term inside via chain rule

Term 1:

$$\therefore J\left(\frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi} + \eta_t \frac{\partial}{\partial \eta}\right)u = J\frac{\partial u}{\partial \tau} + J\xi_t \frac{\partial u}{\partial \xi} + J\eta_t \frac{\partial u}{\partial \eta}$$

$$= \left(\frac{\partial (Ju)}{\partial \tau} - u\frac{\partial J}{\partial \tau}\right) + \frac{\partial (J\xi_t u)}{\partial \xi} - u\frac{\partial (J\xi_t)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - u\frac{\partial (J\eta_t)}{\partial \eta}$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - \left(u\frac{\partial J}{\partial \tau} + u\frac{\partial (J\xi_t)}{\partial \xi} + u\frac{\partial (J\eta_t)}{\partial \eta}\right)$$

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi} + \frac{\partial (J\eta_t u)}{\partial \eta} - \left(\frac{\partial J}{\partial \tau} + \frac{\partial (J\xi_t)}{\partial \xi} + \frac{\partial (J\eta_t)}{\partial \eta}\right)u$$

$= 0$ (geometric conservation law)

$$= \frac{\partial (Ju)}{\partial \tau} + \frac{\partial (J\xi_t u)}{\partial \xi}^{0} + \frac{\partial (J\eta_t u)}{\partial \eta}^{0} = \frac{\partial (Ju)}{\partial \tau}$$

Similar manipulation of term 2, & term 3 (first order spatial derivatives)

term 2:

$$J\left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right)u^2 = \frac{\partial (J\xi_x u^2)}{\partial \xi} + \frac{\partial (J\eta_x u^2)}{\partial \eta} - u^2\left(\frac{\partial (J\xi_x)}{\partial \xi} + \frac{\partial (J\eta_x)}{\partial \eta}\right)$$

$= 0$ (geometric conservation law)

term 3:

$$J\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)uv = \frac{\partial (J\xi_y uv)}{\partial \xi} + \frac{\partial (J\eta_y uv)}{\partial \eta} - uv\left(\frac{\partial (J\xi_y)}{\partial \xi} + \frac{\partial (J\eta_y)}{\partial \eta}\right)$$

$= 0$ (geometric cons. law)

term 4:

$$J\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)\left(\frac{\partial u}{\partial y}\right) = J\xi_y \frac{\partial}{\partial \xi}\underbrace{\left(\frac{\partial u}{\partial y}\right)}_{a} + J\eta_y \frac{\partial}{\partial \eta}\left(\frac{\partial u}{\partial y}\right)$$

$$= J\xi_y \frac{\partial a}{\partial \xi} + J\eta_y \frac{\partial a}{\partial \eta}$$

$$= \frac{\partial (J\xi_y a)}{\partial \xi} - a\frac{\partial J\xi_y}{\partial \xi} + \frac{\partial (J\eta_y a)}{\partial \eta} - a\frac{\partial (J\eta_y)}{\partial \eta}$$

$$= \frac{\partial (J\xi_y a)}{\partial \xi} + \frac{\partial (J\eta_y a)}{\partial \eta} - a\left(\frac{\partial (J\xi_y)}{\partial \xi} + \frac{\partial (J\xi_y)}{\partial \eta}\right)$$

$= 0$ (geometric Cons. law)

$$= \frac{\partial}{\partial \xi}\left(J\xi_y \frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial \eta}\left(J\xi_y \frac{\partial u}{\partial y}\right)$$

Therefore, Final strong conservative form:

$$\frac{\partial (Ju)}{\partial \tau} + \left[\frac{\partial}{\partial \xi}(J\xi_x u) + \frac{\partial}{\partial \eta}(J\eta_x u) \right] + \left[\frac{\partial}{\partial \xi}(J\xi_y v) + \frac{\partial}{\partial \eta}(J\eta_y v) \right] = \frac{\partial}{\partial \xi}\left(J\xi_y \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial \eta}\left(J\xi_y \frac{\partial v}{\partial y} \right)$$

$$\boxed{\frac{\partial (Ju)}{\partial \tau} + \frac{\partial}{\partial \xi}\left(J\xi_x u^2 + J\xi_y uv - J\xi_y \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial \eta}\left(J\eta_x u^2 + J\eta_y uv - J\xi_y \frac{\partial u}{\partial y} \right) = 0}$$

(c) Highlighted terms in blue show the 4 geometric conservation laws (in expression above)

Problem 2

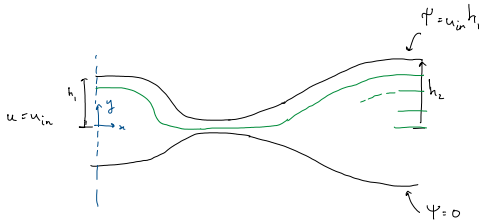
Incompressible potential flow $\nabla^2 \psi = 0$ (ψ: stream function)

(a) Describe the problem

Potential flow is used to model irrotational where the viscosity effects are assumed to be negligible - typically externally flow over solid surfaces or flow at high Reynolds number. In such cases, the thickness of the boundary layer is assumed to be much smaller than the dimen sions of the object.

The given equation is the Laplace equation using stream function. Stream function (ψ) can be defined for any 2D flow such that the stream function is constant along a streamline. Additionally, change in stream function between a pair of streamlines equals the volummetric flowrate between the two streamlines. Finally, lines of constant streamlines are perpendicular to the lines of constant velocity potential (Φ).

(b) Governing equations



$u = \frac{\partial \psi}{\partial y}$ $\quad v = -\frac{\partial \psi}{\partial x}$ 2D, axisymmetric flow ; ψ = stream function (constant along a stream line)

governing equations

$$\therefore \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\downarrow$$

$$\frac{\partial}{\partial x}(-v) + \frac{\partial u}{\partial y} = 0$$

$$\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} = 0$$

BC's :

(Inflow) $u_{in} = \left.\frac{\partial \psi}{\partial y}\right|_{x=0}$ (outflow) $\left.\frac{\partial \psi}{\partial x}\right|_{x=L} = 0$

(walls) $\psi = 0$ (@ curve 1)

$\psi = u_{in} h_1$ (@ curve 2)

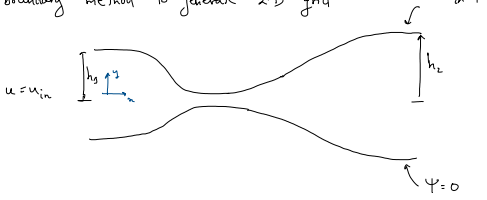
We also use bernoulli's principle to relate pressure & fluid velocity

@ steady state

$$P_{inlet} + \frac{1}{2} \rho u_{in}^2 = \text{constant}$$

@ $x = 0$, $P_{inlet} = P_{atm} = 100{,}000$ Pa

(c) Two-boundary method to generate 2-D grid



$u = \frac{\partial \psi}{\partial y}$ $\quad v = -\frac{\partial \psi}{\partial x}$

$$\int_{(y-h_1)}^{(y+h_1)} u_{in}\, dy = \int_{\psi=0}^{\psi(y)} d\psi$$

$$\psi(y) = u_{in}\left((y+h_1) - (y-h_1)\right) = u_{in}(2h_1)$$

$X_1(\xi,\tau) = x(\xi, \eta=0, \tau)$ $\quad X_2(\xi,\tau) = x(\xi, \eta=1, \tau)$

$Y_1(\xi,\tau) = y(\xi, \eta=0, \tau)$ $\quad Y_2(\xi,\tau) = y(\xi, \eta=1, \tau)$

$$\vec{r_1} = X_1 \vec{i} + Y_1 \vec{j}$$

$$\vec{r_2} = X_2 \vec{i} + Y_2 \vec{j}$$

Hermite interpolation :

$$x(\xi,\eta,\tau) = X_1(\xi,\tau) h_1(\eta) + X_2(\xi,\tau) h_2(\eta) + \frac{\partial x(\xi,\eta=0,\tau)}{\partial \eta} h_3(\eta) + \frac{\partial x(\xi,\eta=1,\tau)}{\partial \eta} h_4(\eta)$$

$$y(\xi,\eta,\tau) = Y_1(\xi,\tau) h_1(\eta) + Y_2(\xi,\tau) h_2(\eta)$$

$$h_1(\eta) = 2\eta^3 - 3\eta^2 + 1$$

$$y(\xi,\eta,\tau) = Y_1(\xi,\tau)\,h_1(\eta) + Y_2(\xi,\tau)\,h_2(\eta)$$

$$h_1(\eta) = 2\eta^3 - 3\eta^2 + 1$$

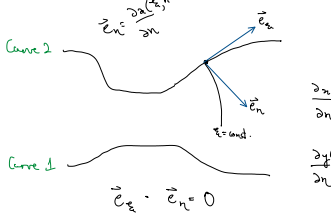
$$h_2(\eta) = -2\eta^3 + 3\eta^2$$

$$h_3(\eta) = \eta^3 - 2\eta^2 + \eta$$

$$h_4(\eta) = \eta^3 - \eta^2$$

$$\vec{e}_\xi = \frac{\partial x_1(\xi,\tau)}{\partial \xi}\vec{i} + \frac{\partial Y_1(\xi,\tau)}{\partial \xi}\vec{j}$$

$$\vec{e}_\eta = \frac{\partial x_2(\xi,\eta=1,\tau)}{\partial \eta}\vec{i} + \frac{\partial y_2(\xi,\eta=1,\tau)}{\partial \eta}\vec{j}$$



$$\vec{e}_\xi \cdot \vec{e}_\eta = 0$$

$$\frac{\partial x_2(\xi,\eta=1,\tau)}{\partial \eta} = -K_2(\xi,\tau)\,\frac{\partial Y_1(\xi,\tau)}{\partial \xi}$$

$$\frac{\partial y_2(\xi,\eta=1,\tau)}{\partial \eta} = +K_2(\xi,\tau)\,\frac{\partial x_1(\xi,\tau)}{\partial \xi}$$

} K-fn. terms

Parametrize the boundaries ; $x = \xi$ $y = f(x) = f(\xi)$

$$\vec{r} = \xi\,\vec{i} + f(\xi)\vec{j}$$

We are given :

Curve 1 : $f_1(x)$

$$\begin{cases} x \in [A,B] \;;\; Y_1 = (D-a_1) - a_1 \cos\left[\dfrac{\pi(x-A)}{(B-A)}\right] \\ x \in (B,C] \;;\; Y_1 = (D-a_2) + a_2 \cos\left[\dfrac{\pi(x-B)}{(C-B)}\right] \end{cases}$$

Curve 2 : $f_2(x)$

$$\begin{cases} x \in [A,B] \;;\; Y_2 = (E+a_1) + a_1 \cos\left[\dfrac{\pi(x-A)}{(B-A)}\right] \\ x \in (B,C] \;;\; Y_2 = (E+a_2) - a_2 \cos\left(\dfrac{\pi(x-B)}{(C-B)}\right) \end{cases}$$

(d) Transform (x,y) to (ξ,η)

$$\nabla^2\psi = 0$$

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} = 0$$

$$\Rightarrow J\left(\xi_x\frac{\partial}{\partial\xi} + \eta_x\frac{\partial}{\partial\eta}\right)\underbrace{\left(\frac{\partial\psi}{\partial x}\right)}_{f} + J\left(\xi_y\frac{\partial}{\partial\xi} + \eta_y\frac{\partial}{\partial\eta}\right)\underbrace{\left(\frac{\partial\psi}{\partial y}\right)}_{g} = 0$$

} strong conservative (times by J^2)

$$\Rightarrow J\xi_x\frac{\partial f}{\partial\xi} + J\eta_x\frac{\partial f}{\partial\eta} + J\xi_y\frac{\partial g}{\partial\xi} + J\eta_y\frac{\partial g}{\partial\eta} = 0$$

$$\Rightarrow \frac{\partial(J\xi_x f)}{\partial\xi} - f\frac{\partial(J\xi_x)}{\partial\xi} + \frac{\partial(\eta_x J f)}{\partial\eta} - f\frac{\partial(J\eta_x)}{\partial\eta} + \frac{\partial(J\xi_y g)}{\partial\xi} - g\frac{\partial(J\xi_y)}{\partial\xi} + \frac{\partial(J\eta_y g)}{\partial\eta} - g\frac{\partial(J\eta_y)}{\partial\eta} = 0$$

$$\Rightarrow \frac{\partial(J\xi_x f)}{\partial\xi} + \frac{\partial(\eta_x J f)}{\partial\eta} + \frac{\partial(J\xi_y g)}{\partial\xi} + \frac{\partial(J\eta_y g)}{\partial\eta} - f\left(\frac{\partial(J\xi_x)}{\partial\xi} + \frac{\partial(J\eta_x)}{\partial\eta}\right) - g\left(\frac{\partial(J\xi_y)}{\partial\xi} + \frac{\partial(J\eta_y)}{\partial\eta}\right) = 0$$

$$\Rightarrow \boxed{\frac{\partial}{\partial\xi}\left(J\xi_x\frac{\partial\psi}{\partial x} + J\xi_y\frac{\partial\psi}{\partial y}\right) + \frac{\partial}{\partial\eta}\left(J\eta_x\frac{\partial\psi}{\partial x} + J\eta_y\frac{\partial\psi}{\partial y}\right) = 0}$$

$$J\xi_x\left(\xi_{1x}\frac{\partial}{\partial e_1} + \eta_y\frac{\partial}{\partial\eta}\right)\psi = \xi_{1x}\left(\frac{\partial(J\xi_x\psi)}{\partial\xi} + \frac{\partial(J\eta_y\psi)}{\partial\eta}\right)$$

$$\therefore \frac{\partial}{\partial\xi}\left(\xi_{1x}\left(\frac{\partial(J\xi_x\psi)}{\partial\xi} + \frac{\partial(J\eta_y\psi)}{\partial\eta}\right) + \xi_y\left(\frac{\partial(J\xi_y\psi)}{\partial\xi} + \frac{\partial(J\eta_y\psi)}{\partial\eta}\right)\right) + \frac{\partial}{\partial\eta}\left(\eta_x\left(\frac{\partial(J\xi_x\psi)}{\partial\xi} + \frac{\partial(J\eta_x\psi)}{\partial\eta}\right) + \eta_y\left(\frac{\partial(J\xi_y\psi)}{\partial\xi} + \frac{\partial(J\eta_y\psi)}{\partial\eta}\right)\right) = 0$$

(e), (f) FDE / FVE's for governing equations + algorithm

$$\left(J^2\xi_x\right)\frac{\partial}{\partial\xi}\left(\xi_x\frac{\partial\psi}{\partial\xi}\right) = \frac{\partial}{\partial\xi}\left[\left(J^2\xi_x^2\right)\frac{\partial\psi}{\partial\xi}\right] - \left(\xi_x\frac{\partial\psi}{\partial\xi}\right)\frac{\partial}{\partial\xi}\left(J^2\xi_x\right)$$

$$\frac{\left(J^2\xi_x^2\right)_{i+\frac12,j}\left(\frac{\partial\psi}{\partial\xi}\right)_{i+\frac12,j} - \left(J^2\xi_x^2\right)_{i-\frac12,j}\left(\frac{\partial\psi}{\partial\xi}\right)_{i-\frac12,j}}{\Delta\xi} - \xi_x\frac{\partial\psi}{\partial\xi}\left[J\frac{\partial}{\partial\xi}(J\xi_x) + \xi_x J\frac{\partial}{\partial\xi}J\right]$$

$$\frac{(J^2\xi_m^2)_{i+\frac{1}{2},j}\left(\frac{\partial\psi}{\partial\xi}\right)_{i+\frac{1}{2},j} - (J^2\xi_x^2)_{i-\frac{1}{2},j}\left(\frac{\partial\psi}{\partial\xi}\right)_{i-\frac{1}{2},j}}{\Delta\xi} \qquad \xi_x\frac{\partial\psi}{\partial\xi}\left[J\frac{\partial}{\partial\xi}(J\xi_x) + \xi_x J\frac{\partial}{\partial\xi}J\right]$$

$$\frac{\psi_{i+1,j}-\psi_{i,j}}{\Delta\xi}$$

$$\frac{\partial}{\partial\xi}(J\xi_x\psi) = \left[(J\xi_x)_{i+\frac{1}{2},j+\frac{1}{2}}\psi_{i+\frac{1}{2},j+\frac{1}{2}} - (Jn)_{i+\frac{1}{2},j-\frac{1}{2}}\psi_{i+\frac{1}{2},j-\frac{1}{2}}\right]/\Delta\xi$$

$$\psi_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{1}{4}(\psi_{i+1,j} + \psi_{i,j} + \psi_{i+1,j+1} + \psi_{i,j+1})$$

$$\vec{A}_\xi = (J\xi_x)\vec{i} + (J\xi_y)\vec{j}$$

$$J = x_\xi y_n - y_\xi x_n \quad ; \quad \xi_x = \frac{y_n}{J} \quad ; \quad \xi_y = -\frac{x_n}{J} \quad ; \quad n_x = -\frac{y_\xi}{J} \quad ; \quad n_y = \frac{x_\xi}{J}$$

} Metric coefficients & Jacobian

$$(\xi_x)_{i,j} = \frac{y_{i,j+1} - y_{i,j-1}}{2\Delta n J_{i,j}} \qquad (\xi_y)_{i,j} = -\frac{x_{i,j+1} - x_{i,j-1}}{2\Delta n J_{i,j}}$$

$$(n_x)_{i,j} = -\frac{y_{i+1,j} - y_{i-1,j}}{2\Delta\xi J_{i,j}} \qquad (n_y)_{i,j} = \frac{x_{i+1,j} - x_{i-1,j}}{2\Delta\xi J_{i,j}}$$

$$J_{i,j} = \left(\frac{x_{i+1,j} - x_{i-1,j}}{2\Delta\xi}\right)\left(\frac{y_{i,j+1} - y_{i,j-1}}{2\Delta n}\right) - \left(\frac{y_{i+1,j} - y_{i-1,j}}{2\Delta\xi}\right)\left(\frac{x_{i,j+1} - x_{i,j-1}}{2\Delta n}\right)$$

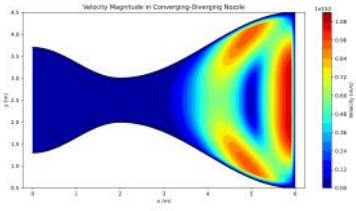
Algorithm:

1) Generate grid via Two-boundary method
   - Use IL, JL = 41 (from lecture) → step sizes of $\Delta\xi = \frac{1}{(IL-1)}$; $\Delta n = \frac{1}{(JL-1)}$
   - Use hermite interpolation
     - → (parameterized curves, computed basis functions, using K factor of 0.08)
     - → Gives grid for X, Y
2) Set initial conditions (u_{in}, P_{in}, ρ)
3) Compute Jacobian & metric coefficients for transformation
4) Finite difference loops for solving for ψ in the transformed domain
5) Inverse transform & using u & v relation to obtain velocity in the physical domain
6) Plotting contour plot

(g), (h) Code + solutions for velocity = 100 m/s ; $P = 100{,}000\ P_a$; $\rho = 1000\ kg/m^3$

See attached document for code
(debugging via LLM software)

The Two-boundary method code was borrowed from the fortran code that was shared in lecture 16, slide 35.



(i) Discuss accuracy of results.

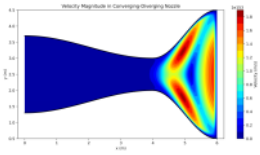
There seems to be a bug in the code which prohibited a physical solution that can be analyzed; however, we can still discuss about the results we expect from a solution to the converging-diverging nozzle

Due the converging geometry, we expect that the velocity will be the highest in the constriction of the nozzle (i.e. for this geometry it should be at x=2); as the fluid moves past the constriction into the diverging part of the nozzle. Since we are assuming incompressible potential flow, the diverging section would decelerate the flow (as the area increases, and velocity decreases). However, this contradicts real compressible flows where you could have supersonic acceleration.

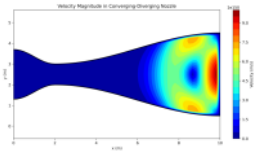
Extra credit: Changing the geometry of the nozzle

Moving the value of B to a higher value of x i.e. B = 4, should simulate the nozzle contraction moving to a later value and hence the expected velocity peak also changes around this constriction. Changing the C parameter will just extend the diverging part of the nozzle which will pretty much make the flow to be converted to flow in a tube (i.e. the peak velocities should decelerate pretty quickly - my code does not show this effect accurately sorry)

Something that would have been also interesting to look at (if my code would have worked) is changing the geometry angles which is known to minimize flow separation and have smooth acceleration to the throat. The throat length can also be varied to stabilize flow before divergence.



Varying the B parameter



Changing the E parameter